# Multi-dimensional Array

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A **multi-dimensional** array in C is an array of arrays. It allows you to store data in a tabular form. (rows and columns or even higher dimensions).

**What is a multi-dimensional Array?**
A multi-dimensional array is declared using multiple brackets.

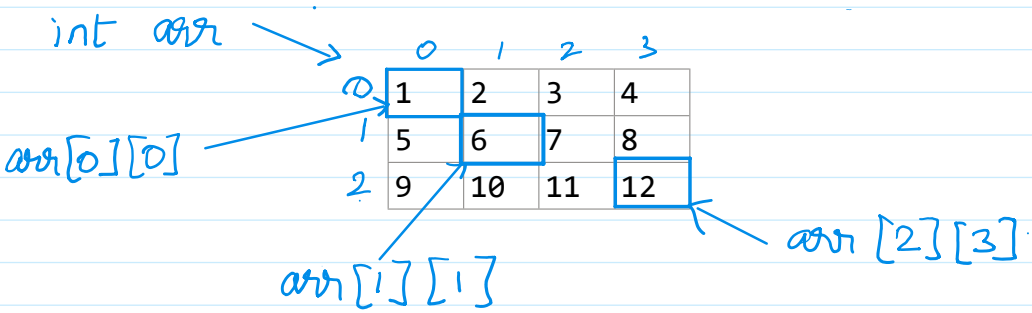
```
data_type array_name[size1][size2]…[sizeN]
```

**Two-Dimensional Array**
A 2D-Array represents a table(matrix)

```
Data_type array_name[rows][columns];
```

For example:

```
int arr[3][4]; //3 rows and 4 columns
```




**Initialization of 2D Array**

**Method-1**

```
int arr[2][3] = {
            {1, 2, 3},
            {4, 5, 6}
            };
```

**Method-2**

```
int arr[][3] = {1, 2, 3, 4, 5, 6}
```

**Accessing Elements:**

```
nameOfArray[row][column]

printf("%d", arr[1][2]); //Output: 6
```

```c
#include<stdio.h>
#define ROWS 3
#define COLUMNS 4
void input(int a[][COLUMNS], const char *msg){
    int i, j;
    printf("%s", msg);
    for(i = 0; i < ROWS; i++){
        for(j = 0; j < COLUMNS; j++){
            printf("Enter values in array [%d][%d]: ", i, j);
            scanf("%d", &a[i][j]);
        }
    }
}
void printArray(int a[][COLUMNS], const char *msg){
    int i, j;
    printf("%s", msg);
    for(i = 0; i < ROWS; i++, printf("\n")){
        for(j = 0; j < COLUMNS; j++){
            printf("\t%d", a[i][j]);
        }
    }
    printf("\nDone");
}
int main(int argc, char const *argv[])
{
    int arr[ROWS][COLUMNS], i, j;
    input(arr, "\nInput values in arr[][]:\n");
    printArray(arr, "\narr[][]:\n");
    return 0;
}
```

**Output:**

```
Input values in arr[][]:
Enter values in array [0][0]: 11
Enter values in array [0][1]: 22
Enter values in array [0][2]: 33
Enter values in array [0][3]: 44
Enter values in array [1][0]: 55
Enter values in array [1][1]: 66
Enter values in array [1][2]: 77
Enter values in array [1][3]: 88
Enter values in array [2][0]: 99
Enter values in array [2][1]: 110
Enter values in array [2][2]: 121
Enter values in array [2][3]: 132

arr[][]:
        11      22      33      44
        55      66      77      88
        99      110     121     132

Done
```

WAP to input two matrix of size mxn and mxn. Take data input from the user and print the matrix and find the sum of two matrix, and print the resultant matrix.

**Homework:**

**Question 1:**

WAP to create a matrix of size NxN and find the sum of each row and each column. Also find the sum of left diagonal and right diagonal.

```
#define N 4
```

| | | | | |
|---|---|---|---|---|
| 10 | -1 | -3 | 9 | = 15 |
| 12 | 2 | -1 | 10 | = 23 |
| 6 | 3 | 12 | 9 | = 30 |
| 4 | 8 | 6 | 10 | = 28 |
| 32 | 12 | 14 | 38 | = 34 ← left diagonal |

15 ↑ right diagonal

**Question 2:**

WAP to create two matrix of size MXN and find the product of two matrix. Print the matrix along with product matrix.